Name: ____________________

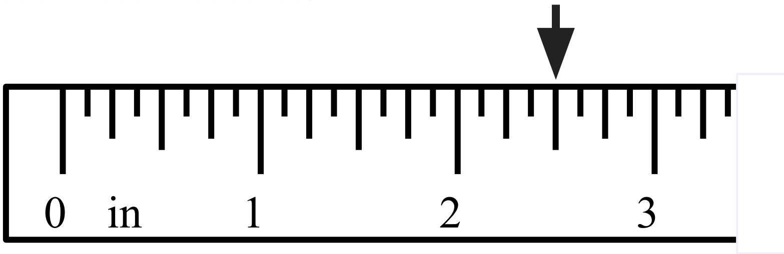
Worksheet

Engineer

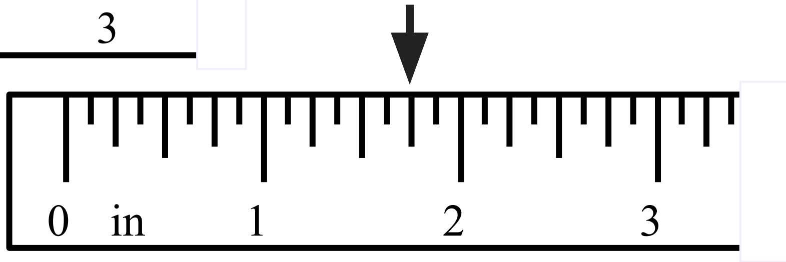
Task A: Precision in engineering

1) Read these measures.

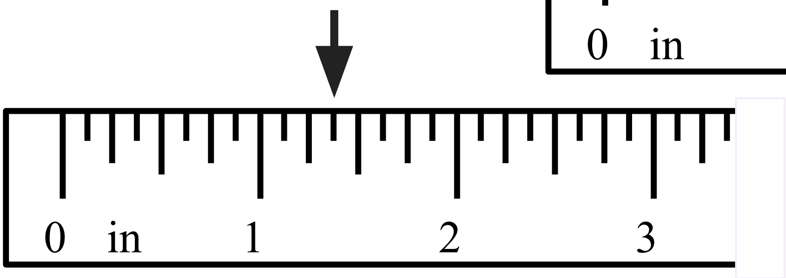
a)



b)

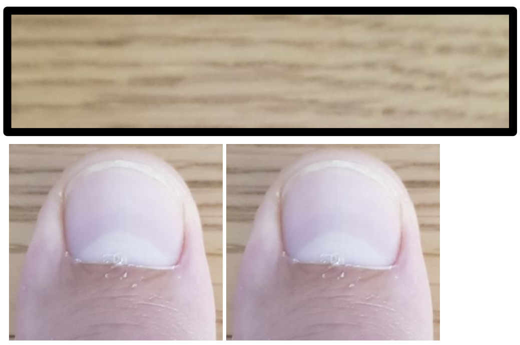


c)



2) Use a rule of thumb to measure out the below:

a) One and three eighths of an inch.



b) Two and three sixteenths of an inch.

3) a) How many millimetres in a centimetre?

One centimetre	$\frac{1}{100}$ m	1×10^{-2} m
One millimetre	$\frac{1}{1000}$ m	1×10^{-3} m
One micrometre	$\frac{1}{1\,000\,000}$ m	1×10^{-6} m

b) How many micrometres in a millimetre?

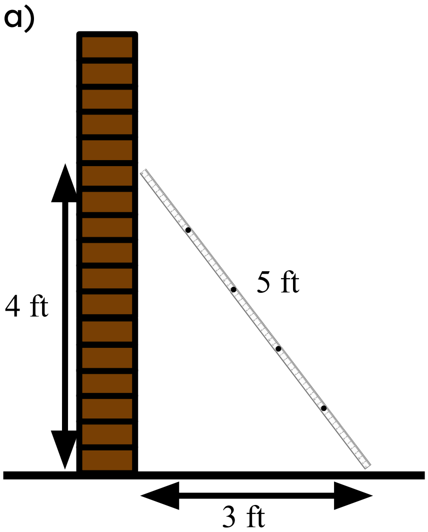
c) How many micrometres in a centimetre?

d) Engineers can laser cut to a precision of 2.5×10^{-5} m
What fraction of a centimetre is this?

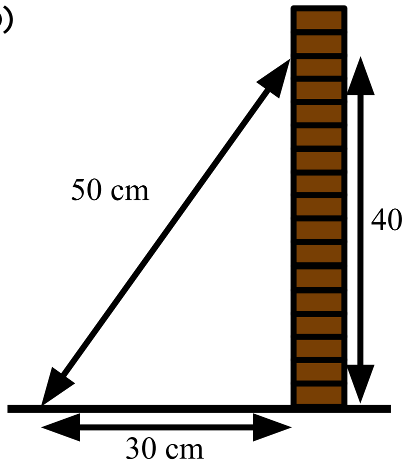
Task B: Other applications of maths

1) Show whether each of these walls is upright.

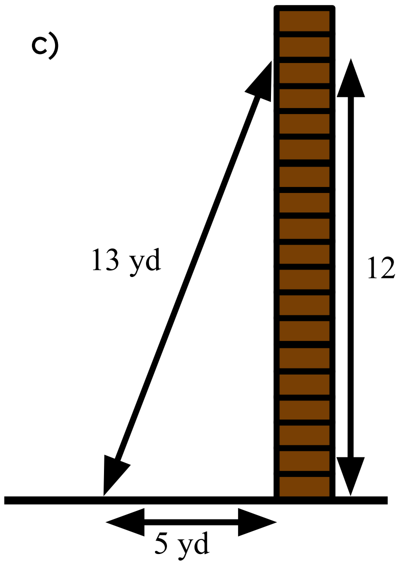
a)



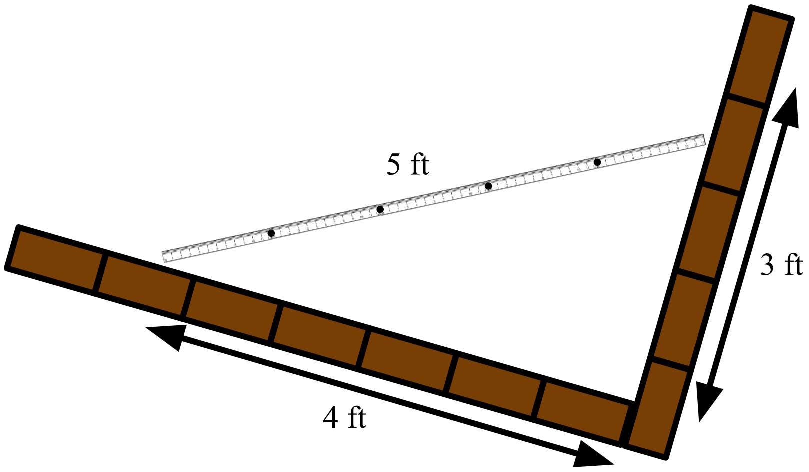
b)



c)



2) Explain how we know this corner is not 'right'.



3) Engineers today use Pythagoras to calculate lengths. Find all the missing lengths of the girders on this bridge. The design is symmetrical.